

ensure compliance with standards and product specifications and initiate design changes as per the requirement.

If you are comfortable with software, you can begin as an embedded system programmer. This role will engage you in activities like analysis and optimisation of embedded software for the targeted real-time operating system (RTOS). Your tasks may include developing installable and built-in device drivers, kernel modifications and embedded application.

If you would like to explore the application possibilities in the field, give a thought to becoming a protocol engineer. You will be assigned tasks like development, integration and testing of various protocols within an embedded firmware stack used in devices.

However, keep in mind that you are expected to be an industry-ready product. "There is a shortage of industry-ready talent in the embedded system field, which means industry-specific talent should be employed by imparting trainings to freshers," says Patel.

Who is hiring?

Almost all industries are increasing the use of embedded systems for increased sophistication in the areas of networking, automation and control. As a result, software, telecom and electronics companies are heavily diversifying into this field. The demand is maximum for professionals having 2-5 years of experience. Entry-level hiring is around 20 per cent of the total talent acquisition.

Most semiconductor manufacturers are involved in VLSI chip design and embedded software development. Indian companies are developing embedded products in consumer, industrial, telecom, automotive, computing and medical spaces, for local as well as foreign markets. Transnational global companies like Cisco, Juniper, Honeywell, GE, Bosch, Delphi, Visteon and Continental have set up their embedded development centres in India. In addition, semiconductor and design services companies in India like Wipro, HCL, Texas Instruments, Freescale,

Cadence, Mindtree, L&T, TCS, Tata Elxsi and KPIT Cummins are carrying out embedded software and hardware product development in India for global accounts.

Though the major recruiters are big players, you should also take into account the small and medium enterprises (SMEs). In fact, these are responsible for more than 50 per cent recruitment. Naveen Kumar, CEO, Emtech Foundation (an embedded system training institute), informs that SMEs are located in Delhi, Noida, Gurgaon, Pune, Bengaluru, Hyderabad, Chennai and even smaller cities like Nagpur, Mysore and Kanpur.

What's on offer?

Due to the huge gap between demand and supply of quality human resource, an experienced embedded system engineer may be the most sought-after

depends on the nature of organisation and competency level and academic background of the candidate. The ability to handle multitasking jobs may also be a defining factor.

After successful completion of the training period, an engineer's career graph grows from the junior level towards a project leader, project champion or project manager position. In addition to gaining technical expertise, it is always advisable to explore the business processes as that can accelerate the climb to the top.

Kumar opines that in an SME one can manage a growth of 20-30 per cent up to 100 per cent after gaining an experience of one or two years, depending on whether one decides to stick to the same firm or switch jobs. He also points out the many onsite opportunities that exist in USA, Europe, Singapore and Japan. You can explore

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professional in the electronics industry. "I believe salary is never a constraint for the right candidate. Typically, an embedded fresher starts at the same level as other engineers getting their first job in IT companies. However, embedded professionals are required to be more of solution seekers and providers. Due to these roles, they get to develop their skills faster and their rise on the corporate ladder too is faster," opines Krishna.

Let us check what is on offer. According to Patel, the average salary for a fresher in the embedded system field can range anywhere from Rs 10,000 to Rs 20,000 per month. Well-funded companies are prepared to offer upwards of Rs 25,000. Analysis of previous year's industry feedback reveals figures ranging between Rs 1.2 million and Rs 2 million per annum for candidates with four to five years of experience. The compensation

those after gaining experience of three to five years.

Know the trivia

So now you want to know the basic criteria for entry, isn't it? Diploma holders, engineering as well as science graduates and postgraduates, and even doctorates with background in electronics/electrical engineering or computer science may try their luck here. Primarily, embedded system development requires skills in electronics and computer science disciplines. The specific skills required are algorithms and programming, basic maths, computer organisation, electrical networks and digital circuits, control systems, digital signal processing (DSP) and basics of machine vision, real-time systems (including RTOS), programming of microprocessors and microcontrollers, and lab hardware.

"Depending upon the specialisa-